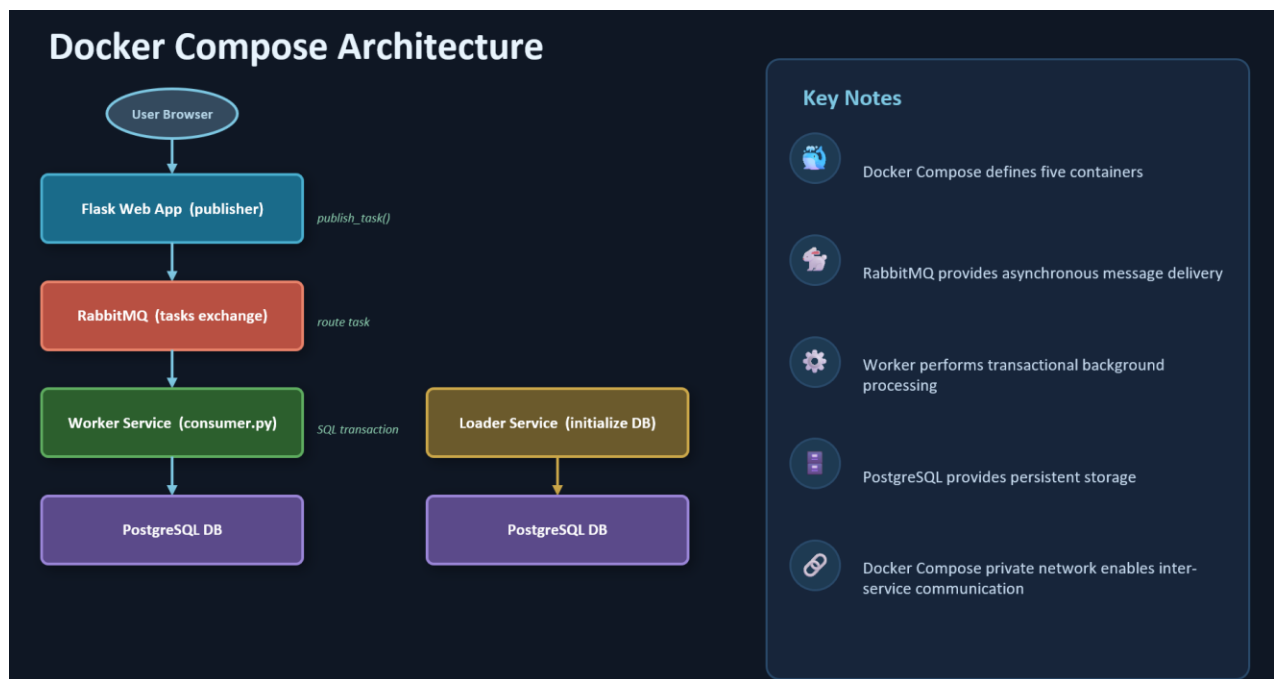


Module 6 Deployment Report

1. System Architecture

This project implements an asynchronous microservice architecture using Docker Compose, Flask, RabbitMQ, PostgreSQL, and a background worker service.

Figure 1. Docker Compose architecture showing the Flask web service, RabbitMQ message broker, worker service, PostgreSQL database, and loader service.



2. Service Roles

- Flask Web Service: publishes tasks to RabbitMQ.
- RabbitMQ: provides asynchronous message delivery.
- Worker Service: consumes messages and executes background tasks.
- PostgreSQL: stores applicant data and analytics.
- Loader Service: initializes the database schema and seed data.

3. RabbitMQ Message Flow

Flask publishes tasks to a durable RabbitMQ exchange. Messages are routed to the durable tasks_q queue. The worker consumes one message at a time (prefetch_count=1), performs database transactions, commits on success, and acknowledges the message.

4. Database Initialization

The loader service initializes the PostgreSQL database before the web and worker services begin processing tasks.

5. Verification Results

- Pytest: 57 tests passed
- Coverage: 100%
- Pylint: 10/10
- Docker Compose deployment: successful
- RabbitMQ messaging: successful
- Docker Hub publishing: successful
- End-to-end asynchronous workflow: verified

Figure 2. Pytest and Pylint verification results

```
===== tests coverage =====
_____ coverage: platform win32, python 3.14.5-final-0 _____

Name                               Stmts  Miss  Cover   Missing
-----
src\__init__.py                     0      0   100%
src\db\load_data.py                 73      0   100%
src\web\app\app.py                  33      0   100%
src\web\publisher.py                22      0   100%
src\worker\etl\clean.py             89      0   100%
src\worker\etl\pull_new_data.py     95      0   100%
src\worker\etl\query_data.py        55      0   100%
src\worker\etl\scrape.py           102      0   100%
-----
TOTAL                               469      0   100%
Required test coverage of 100% reached. Total coverage: 100.00%
===== 57 passed, 5 warnings in 4.37s =====

-----
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)
```

6. Screenshots

Figure 3. Running Flask web application (localhost:8080).

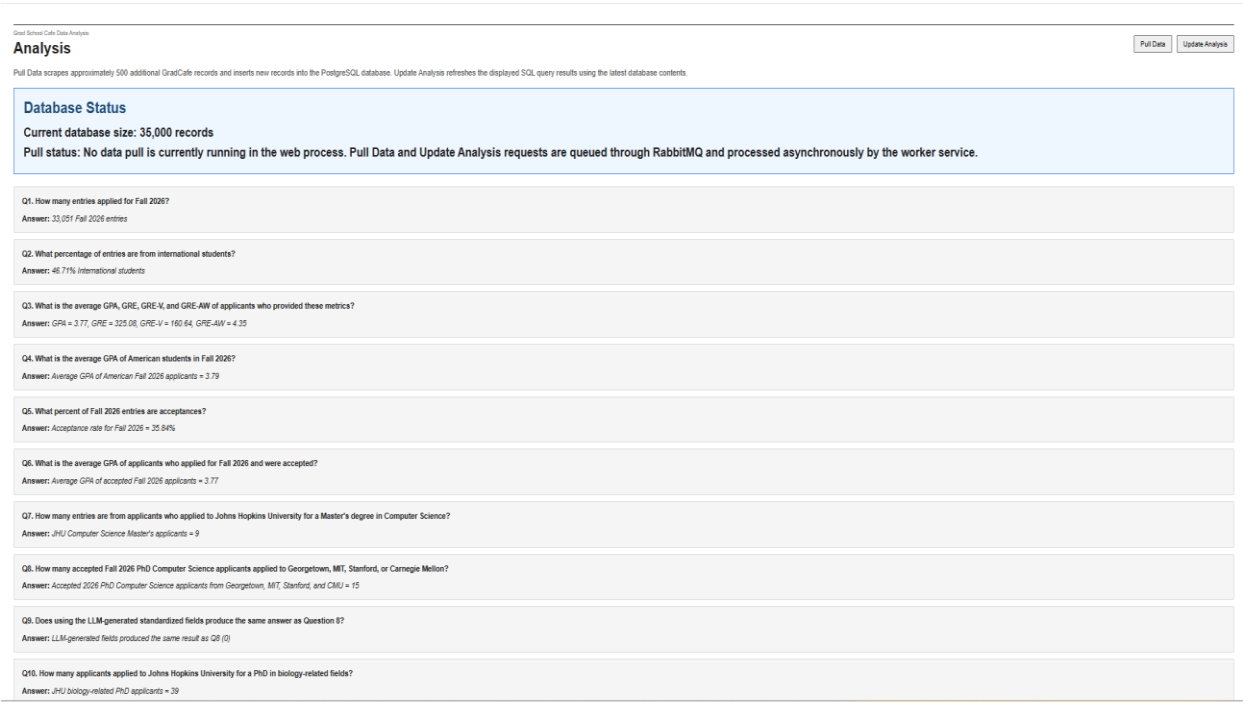


Figure 4. RabbitMQ durable queue configuration showing tasks_q queue status and durable messaging.

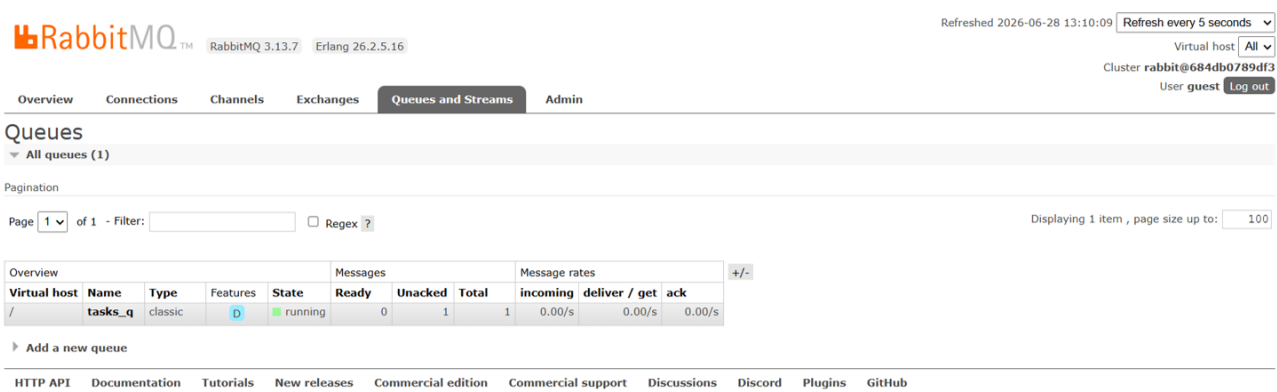


Figure 5. Docker Compose deployment status and worker execution log showing successful asynchronous task processing, PostgreSQL insertion, and RabbitMQ message acknowledgement.

```
zhk11@HKZ_computer MINGW64 ~/Documents/jhu_software_concepts/module_6 (main)
$ docker compose ps
NAME                                IMAGE                                COMMAND                                SERVICE    CREATED      STATUS      PORTS
module_6-db-1                       postgres:16                         "docker-entrypoint.s..."          db         11 minutes ago Up 11 minutes (healthy) 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
module_6-rabbitmq-1                 rabbitmq:3-management              "docker-entrypoint.s..."          rabbitmq    11 minutes ago Up 11 minutes (healthy) 0.0.0.0:5672->5672/tcp, [::]:5672->5672/tcp, 0.0.0.0:15672->15672/tcp, [::]:15672->15672/tcp
module_6-web-1                      module_6-web                        "python -m src.web.r..."          web         11 minutes ago Up 11 minutes          0.0.0.0:8080->5000/tcp, [::]:8080->5000/tcp
module_6-worker-1                  module_6-worker                    "python -m src.worke..."          worker      11 minutes ago Up 11 minutes

zhk11@HKZ_computer MINGW64 ~/Documents/jhu_software_concepts/module_6 (main)
$ docker compose logs worker --tail=30
worker-1 | Fetching page 1814...
worker-1 | Collected 300 records
worker-1 | Fetching page 1815...
worker-1 | Collected 320 records
worker-1 | Fetching page 1816...
worker-1 | Collected 340 records
worker-1 | Fetching page 1817...
worker-1 | Collected 360 records
worker-1 | Fetching page 1818...
worker-1 | Collected 380 records
worker-1 | Fetching page 1819...
worker-1 | Collected 400 records
worker-1 | Fetching page 1820...
worker-1 | Collected 420 records
worker-1 | Fetching page 1821...
worker-1 | Collected 440 records
worker-1 | Fetching page 1822...
worker-1 | Collected 460 records
worker-1 | Fetching page 1823...
worker-1 | Collected 480 records
worker-1 | Fetching page 1824...
worker-1 | Collected 500 records
worker-1 | Saved 500 records to additional_applicant_data.json
worker-1 | Saved 500 records to additional_cleaned_applicant_data.json
worker-1 | Inserted 500 new records into PostgreSQL.
worker-1 | Skipped 0 duplicate or invalid records.
worker-1 | Pull Data completed successfully.
worker-1 | Task completed and acknowledged: scrape_new_data
worker-1 | Handling recompute_analytics with payload={}
worker-1 | Task completed and acknowledged: recompute_analytics

zhk11@HKZ_computer MINGW64 ~/Documents/jhu_software_concepts/module_6 (main)
$
```